

Spencer Hum

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EDUCATION

Queen's University

Bachelor of Computing (Honours) in Software Design

Sept. 2022 – Expected Apr. 2027

EXPERIENCE

Yupi.ca

Software Engineer Intern

May 2025 – Apr. 2026

- Developed a data processing pipeline using **Django ORM and batch-processing SQL scripts** to categorize operational expenses and automate daily reconciliation of trading and payment fees
- Implemented **Celery with RabbitMQ** to asynchronously validate and reconcile high-volume trade jobs, reducing processing latency and improving system throughput during peak settlement periods
- Minimized backend API traffic by caching high-frequency GET requests through **Akamai CDN using Terraform**
- Deployed a fiscal-year financial dashboard using **Django models and SQL queries** to analyze product sales

Canada Computers & Electronics

Software Developer Intern

Sept. 2024 – Dec. 2024

- Developed and designed the front end of a B2B e-commerce platform using **React, TypeScript, and Tailwind CSS**, building scalable UI components and ensuring responsiveness across modern browsers
- Implemented **secure user authentication, role-based access control, and account management features**, enabling clients to manage their own accounts without external support
- Worked directly with stakeholders to define requirements and iterate on UX, resulting in a more **self-service-oriented client experience** and reduced support overhead

PROJECTS

Quantitative Trading Framework | *Python, QuantLib, Pandas, Redis*

- Built an options pricing engine in Python, leveraging the Black-Scholes model and QuantLib
- Secured an 8% increase in portfolio returns by automating trading strategies in backtesting simulations
- Reduced portfolio risk by 12% by performing Monte Carlo emulations to identify potential market downturns
- Integrated real-time market data using Redis to enable trade adjustments based on Black-Scholes valuations

QUMarketplace: Marketplace Web App | *MongoDB, ExpressJS, TypeScript, JavaScript, React*

- Developed a full-stack online marketplace platform using React, TypeScript, Express.js, and MongoDB, replicating core Facebook Marketplace features including listings, messaging, and user profiles
- Implemented secure login and registration flows using JWT and bcrypt, enabling encrypted sessionless authentication for users with role-based access control

CookKeeper: Recipe Library | *PostgreSQL, ExpressJS, TypeScript, JavaScript, React*

- Designed a full-stack web application to manage personal recipes with CRUD functionalities for recipe entries
- Designed back-end routes and middleware with ExpressJS to ensure smooth data flow between the client and database, with thorough endpoint validation using Postman

SKILLS

Certifications: AWS Certified Cloud Practitioner

Languages: Python, JavaScript, Typescript, HTML/CSS, ExpressJS, Django

Tools: React, Git, Redis, NodeJS, MongoDB, PostgreSQL, MySQL, Celery Beat, RabbitMQ, AWS, Akamai, Terraform